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Method for dispensing drops of different volumes

Abstract

A method that allows a user to dispense a desired volume of solution from an acoustic dispensing apparatus by allowing the user to select the drop volume to be dispensed. A typical drop volume is in the range of one to twenty-five nanoliters. The method comprises the steps of creating two or more burst curves that give the relationship between liquid level and burst value, using data from the burst curves to create two or more calibration functions, and using data from the calibration functions to create a dispensing data set that is used to set the burst parameter required to dispense the selected drop volume.

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Background/Summary

CROSS-REFERENCES TO RELATED APPLICATIONS (1) This application is a divisional of U.S. application Ser. No. 15/886,744, filed on Feb. 1, 2018, which is incorporated herein by reference in its entirety.

BACKGROUND OF THE INVENTION

(1) Acoustic dispensing is a well-known method for dispensing very small volumes of liquid, for example in the range of one nanoliter to one microliter. Generally, multiple drops (sometimes called droplets) having a fixed volume are ejected from an acoustic dispensing apparatus to yield

the total volume of liquid that is desired. This methodology is used because the acoustic dispensing apparatus must be carefully calibrated to dispense a specific drop volume, making it time consuming to change the volume of the drop. The inability to freely select the volume of a drop creates several problems. For example, it limits the final dispense volume to multiples of the selected drop volume, and it results in a longer dispense time if the size of the drop is small. Nonetheless, this is the current state of acoustic dispensing using technology such as that discussed below.

(2) FIG. 1 illustrates an acoustic drop dispensing apparatus **10** known in the prior art. Apparatuses of this type are capable of dispensing drops of liquids having volumes as small as approximately one hundred picoliters, and are particularly useful in the biotechnology and biopharmaceutical fields. A representative acoustic drop dispensing apparatus is described in U.S. Pat. No. 6,863,362 which is incorporated herein by reference.

(3) In the apparatus **10**, an acoustic wave emitter **14** (such as a piezoelectric crystal) is in electrical communication with a computer **18**. During operation the acoustic wave emitter **14** generates an acoustic wave or beam **20** that can be propagated through an optional wave channel **24**. The acoustic wave can be focused by a lens **28** prior to propagating through a coupling medium **32** to optimize the energy of the acoustic wave or beam **20** upon the liquid/air interface (free surface) of a source liquid **40**. The assembly comprised of the acoustic wave emitter **14**, the wave channel **24** and the lens **28** is referred to as an acoustic emitter assembly **29**. The acoustic wave **20** is propagated through the coupling medium **32** after which the wave is transmitted through a source liquid containment structure **44** where the wave comes to focus at or near the surface of the pool of source liquid **40**, thereby causing a drop **60** of the source liquid **40** to be dispensed from the surface of the pool.

(4) Examples of source liquid containment structures **44** include single and multi-well wellplates commonly used in molecular biology applications, capillaries (e.g., capillary arrays), and the like. However, other containers or structures may be used to hold the liquid **40** to be dispensed or ejected. A typical wellplate comprises a matrix (rows and columns) of individual wells **46**. Typical commercially available wellplates have 96, 384, 1536 or 3456 individual wells. The source liquid **40** may be contained in some or all of these wells **46** and the composition of the source liquid in individual wells may differ from well to well (i.e. there can be multiple source liquids **40**). Furthermore, the volume of source liquid in the individual wells may differ from well to well. The volume of source liquid in an individual well is derived from the liquid level and well geometry.

(5) Optimally, to dispense one or more drops from one of the individual wells **46**, the well **46** must be positioned over the acoustic wave emitter **14**. To accomplish this, the source fluid containment structure **44** is detachably affixed to a gripper **49**. The gripper **49** is controlled by an actuator mechanism **50** which contains a horizontal actuator **54** for moving the containment structure **44** in the horizontal (x and y) directions. A vertical actuator **58** moves the acoustic wave emitter **14** and wave channel **24** in the vertical (z) direction. The actuator **50** is typically in communication with computer **18** which controls the movement of the containment structure **44** to select a source liquid **40** or to adjust focusing of the acoustic wave or beam **20** at or near the surface of the source liquid **40**. The computer may have implemented thereon various algorithms to adjust the focal position and energy of the acoustic wave emitter as well as control and manage the location of the acoustic wave emitter relative to a source fluid present in or on a source fluid containment structure.

(6) Accordingly, the apparatus **10** may be used to cause one or more drops **60** of the source liquid **40** to be dispensed from the containment structure **44** and towards a target substrate **70**, as is described in U.S. Pat. No. 6,863,362. The target substrate **70** may be a multi-well wellplate like the source fluid containment structure **44**, or may be some other type of medium. Generally, one or more horizontal actuators **59** are provided for moving the target substrate **70** in the horizontal (x and y) directions. A typical wellplate that could be used as the target substrate **70** may have 96, 384, 1536 or 3456 individual target wells **74**, or some other number of target wells. FIG. 2

illustrates the target wells **74** in a wellplate used as the target substrate **70**.

(7) In many cases, a piezoelectric transducer is employed as an acoustic wave emitter **14**. For example, the piezoelectric transducer may comprise a flat thin piezoelectric element, which is constructed between a pair of thin film electrode plates. As is understood by those of skill in the art, when a high frequency and appropriate magnitude voltage is applied across the thin film electrode plates of a piezoelectric transducer, radio frequency energy will cause the piezoelectric element to be excited into a thickness mode oscillation. The resultant oscillation of the piezoelectric element generates a slightly diverging acoustic beam of acoustic waves. By directing the wave or beam onto an appropriate lens having a defined radius of curvature (e.g., a spherical lens, or the like), the acoustic beam can be brought to focus at a desired point.

(8) Generally, a computer sends an analog voltage pulse to the piezoelectric transducer by an electrical wire **78**. The electronics can control the magnitude and duration of the analog voltage pulses, and the frequency at which the pulses are sent to the piezoelectric transducer. Each voltage pulse causes the generation of an acoustic wave from the piezoelectric transducer, which in turn is propagated through a coupling medium and into or through the source fluid thereby impinging on the surface of the source fluid. A series of cycles of acoustic waves and one “off” period after the generation of the acoustic waves (corresponding to an interval between voltage pulses) is referred to as one “burst.”

(9) A problem encountered in using acoustic drop dispensing systems, such as the apparatus **10**, is that it is difficult to precisely control the volume of the drops dispensed from the apparatus. In large part, this is because many parameters associated with the source liquid, such as chemical composition, viscosity, temperature, speed of sound in the liquid, etc., affect the size (volume) of the drop. Furthermore, the liquid level of the source liquid in the well **46** also affects the size (volume) of the drop. Additionally, other factors, such as the geometry of the source well (e.g. well shape, well bottom thickness, etc.) or the manufacturing variability of the acoustic emitter assembly **29**, can influence the size of the drop. To deal with this problem, the acoustic drop dispensing apparatus **10** needs to be calibrated so that uniform drop volume can be achieved. A method for calibrating the apparatus **10** is described in U.S. Pat. No. 7,661,289 which is incorporated herein by reference.

(10) As was mentioned previously, the inability to freely select the volume of a drop to be dispensed limits the final dispense volume to multiples of the selected drop volume, and results in a longer dispense time if the size of the drop is small. What is needed is the ability to select and dispense drops of any volume within a reasonable range of drop volumes. This would allow the drop volume to be optimized based on the final volume of source solution to be dispensed. In other words, fewer drops of larger volume could be used to accomplish the dispense volume, and the user could choose the dispense volume that is desired.

BRIEF SUMMARY OF THE INVENTION

(11) Briefly, the present invention is an acoustic dispensing method that allows the user to select the final total volume of solution to be dispensed. The method creates the most efficient drop volume calibration needed for dispensing by the acoustic dispensing apparatus. The user can also manually select the drop volume that allows the dispense time to be minimized because the volume of the drops can be chosen to minimize the number of drops that need to be dispensed to yield the final total volume.

(12) The method comprises the steps of creating two or more burst curves that give the relationship between liquid level and burst value, using data from the burst curves to create two or more calibration functions, and using data from the calibration functions to create a dispensing data set that is used to set the burst parameter required to dispense the selected drop volume. In a typical procedure, the user determines the number of drops needed to dispense the desired volume of solution, calculates the required drop volume, uses the liquid level of the solution to select the burst parameter from the dispensing data set, and dispenses the drops.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic diagram of an acoustic drop dispensing apparatus of the prior art;
- (2) FIG. 2 is a top view of a wellplate of the prior art;
- (3) FIG. 3 is a flow chart according to the present invention;
- (4) FIG. 4 is a plurality of burst curves;
- (5) FIG. 5 is a graph of a calibration function according to the present invention;
- (6) FIG. 6 is a graph of the slopes of a plurality of calibration functions according to the present invention;
- (7) FIG. 7 is a graph of the intercepts of a plurality of calibration functions according to the present invention;
- (8) FIG. 8 is a cross sectional view of a well in a wellplate;
- (9) FIG. 9 is a graph illustrating the drop volume required to dispense a requested volume; and
- (10) FIG. 10 is a flow chart illustrating a method of using the present invention.

DETAILED DESCRIPTION OF THE INVENTION

- (11) The present invention is a method that allows a user to select any drop volume within a specified range for dispensing by the acoustic dispensing apparatus 10. Among other things, the ability to select the drop volume allows the dispense time to be minimized because the volume of the drops can be chosen to minimize the number of drops needed to yield the total volume that needs to be dispensed. It also gives the user more freedom in selecting the total volume to be dispensed, because the dispensing process is no longer limited to one or two drop sizes.
- (12) The ability to select the drop volume is accomplished through a procedure that creates two or more calibration functions that relate drop volume to burst over a range of liquid levels in a containment structure, such as a well in a wellplate. In the preferred embodiment, the calibration function is generated through a multiple part method, whose endpoint is a dispensing data set allows the burst needed to produce the desired drop volume to be set. FIG. 3 is a flow chart that summarizes the sequence of steps used in generating the dispensing data set that allows a user to choose any drop volume within a selected range for dispensing by the acoustic dispensing apparatus 10.
- (13) Step 130 in FIG. 3 illustrates the first step in the procedure for generating the calibration functions. In step 130, a series of burst curves are derived for a range of drop volumes. A burst curve is a plot of liquid level versus burst value for a fixed drop volume. In a representative calibration procedure, the data for a plurality of burst curves is collected for a plurality of drop volumes. For example, FIG. 4 illustrates a set of seven burst curves for seven different drop volumes, such as 2, 4, 6, 7, 8, 9, and 10 nanoliters. In the preferred embodiment, the burst curves are generated as part of the calibration procedure used in a commercially available acoustic dispensing apparatus, and are discussed later in this patent application.
- (14) In step 136, for a fixed liquid level value (e.g. 3.01 mm), drop volume versus burst parameter is plotted for the drop volumes used in the first step 130 (seven drop volumes in this example). In other words, by using the burst curves from FIG. 4 and step 130, the burst value for each of the drop volumes is extracted from the burst curve at the fixed liquid level, and plotted versus drop volume, as is illustrated in FIG. 5. Step 142 illustrates that the data from step 136 is fitted to a function, referred to as the calibration function, that can be used to determine the burst value needed to produce a selected drop volume of any size at the specified liquid level.
- (15) In the preferred embodiment, the function that results from step 142 is the equation for a straight line relating drop volume to burst value for a given liquid level. Therefore, the slope and intercept of this function can be used to calculate the burst value needed to produce any selected drop volume at the specified liquid level. The curve fitting process can be accomplished by several

methods. For example, commercial software like Microsoft's Excel spreadsheet can be used, as can National Instruments' LabVIEW math functions software. Alternatively, a curve fitting routine, like a least squares fitting routine, can be custom written and compiled in a computer language like C++.

(16) FIG. 5 illustrates the graph generated in steps **136** and **142** using the data from FIG. 4 at the liquid level of 3.01 mm, and shows the calibration function **194** as a straight line. In other embodiments, the calibration function may not be a straight line (i.e. the calibration function might be a curve), and a higher degree polynomial would be required for an acceptable the curve fit.

(17) Step **148** indicates that in the preferred embodiment, a plurality of calibration functions is needed for a plurality of liquid levels (e.g. **36** calibration curves for 36 liquid levels). The plurality of calibration functions is generated by repeating steps **136** and **142** for the plurality of liquid levels, thereby yielding a plurality of different calibration functions analogous to the calibration function **194** shown in FIG. 5.

(18) Step **154** indicates that each of the plurality of different calibration functions has a slope and intercept (or other coefficients) that can be used to calculate the burst needed to produce any selected drop volume over the range of liquid levels. A lookup table (called the dispensing data set) is created in step **154** for storing the slope and intercept and any other relevant information as a function of liquid level.

(19) FIGS. 6 and 7 show the results of Step **154** in graphical form. FIG. 6 illustrates the individual values for the slopes of the plurality of calibration functions plotted over the range of liquid levels from 0.05 to 6.54 millimeters, and FIG. 7 illustrates the individual values for the intercepts of the plurality of calibration functions plotted over this range of liquid levels. The data contained in FIGS. 6 and 7 can be used to calculate the burst value needed to produce any selected drop volume over the range of liquid levels, as is explained later.

(20) In a preferred embodiment, the data in FIG. 4 was generated using a Gen 5 ATS acoustic dispenser from EDC Biosystems of Fremont, California. The Gen 5 ATS acoustic dispenser is similar to the acoustic dispensing apparatus **10** shown in FIG. 1, and in the discussion below, the components of the acoustic dispenser **10** shown in FIG. 1 are used to describe the Gen 5 ATS acoustic dispenser.

(21) The seven burst curves shown in FIG. 4 show what burst value must be used in the acoustic dispenser **10** to dispense the specified drop size (volume) over a range of liquid levels. A burst is a series of acoustic waves followed by a period of rest, such as an "off" period after the generation of the acoustic waves. The off period corresponds to an interval between voltage pulses applied to the acoustic wave emitter **14** that cause the acoustic waves to be emitted. Therefore, a series of bursts is proportional to the length of time that an acoustic signal is applied to a source liquid. In other words, the amount of energy being applied to the surface of the liquid is proportional to both the strength of the acoustic wave and the length of time that those acoustic waves are being applied. In this application, the terms burst, burst value, and burst parameter are used interchangeably.

(22) In order to generate a burst curve, the apparatus **10** must be calibrated to determine what burst will yield a specific drop volume at various liquid levels in the source well. In the preferred embodiment, a method for generating burst curves such as the one described in U.S. Pat. No. 7,661,289, is used. In other embodiments, other methods could be used. In general, the calibration procedure involves using the apparatus **10** to dispense drops of a solution containing a dye, such as a fluorescent dye, into target wells, and then calculating the drop volume that was dispensed by comparison to a standard having a known concentration of the fluorescent dye. This process is repeated for a plurality of liquid levels, and then the data is processed to yield the burst curve showing what burst is required to produce a drop of a given volume as a function of liquid level (i.e. as a function of the height of the source liquid in a well or some other container).

(23) Once an acceptable burst curve is obtained, it can be further processed to yield a fine tuned burst curve. Fine tuning is done by selecting the burst settings from a burst curve for a particular

drop volume, and then using the settings to dispense a dye solution, such as fluorescein dye in a DMSO solution, onto a target plate using the apparatus **10**. The fluorescent counts versus liquid level are then plotted and compared to the fluorescent counts expected based on the selected drop size. The burst values for each liquid level are then changed until the fluorescent counts are roughly uniform over the range of liquid levels, indicating that a uniform (and accurate) drop size is being dispensed at each liquid level. A calibration is considered finely tuned when the relative standard deviation is less than 5% along the range of liquid levels. The mean value of the data is used as the actual drop volume dispensed in the process.

(24) FIG. **8** illustrates that the liquid level “L” of the source liquid **40** in the source well **46** is the height of the free surface of the liquid **40** above the bottom of the well **46**. Generally, L is the distance between the lowest part of the meniscus **84** of the liquid **40**, and a well bottom surface **86** of the well **46** that is in contact with the liquid **40**. However, other reference points could be used as the liquid level.

(25) FIG. **8** also illustrates that the wellplate **44** has a thickness “T” underneath the well bottom surface **86**. The wellplate **44** has a wellplate bottom surface **88**. In the preferred embodiment, liquid level is measured by the acoustic dispenser **10**, such as by measuring the time it takes for an acoustic wave to make a round trip from the acoustic wave emitter **14** to the surface of the source liquid **40** (i.e. the meniscus **84**), called t1, and subtracting out the time it takes for an acoustic wave to make a round trip to the bottom surface **86** of the well, called t2. The liquid level (LL) is then calculated using a calculation such as $LL = v(t1 - t2)/2$, where v is the speed of sound in the liquid **40**, as is explained in U.S. Pat. No. 7,661,289. However, other methods of measuring liquid level could be used.

(26) Referring to FIG. **4**, seven finely-tuned burst curves are shown for seven different drop volumes of a 90% DMSO/10% water/100 μ M fluorescein solution. These curves are labeled 164, 168, 172, 176, 180, 184, and 188, and correspond to drop volumes of 2 nanoliters, 4 nl, 6 nl, 7 nl, 8 nl, 9 nl, and 10 nl, respectively. In practice, since this calibration procedure is implemented in software, the data for the burst curves are stored as a calibration files in electronic memory (usually on a hard disk and in RAM).

(27) A calibration file is created for each drop volume (i.e. seven calibration files in this example), and each calibration file comprises a look up table that lists a variety of parameters required to dispense the given drop volume. These parameters include the drop volume, focus, voltage, and burst for each of thirty-six liquid levels. In the preferred embodiment, the focus and voltage are held constant, so only the burst varies with liquid level. In other embodiments, the focus and voltage could be varied, and other parameters could be included. The calibration files are referred to as burst curve data sets in other parts of this application.

(28) FIG. **5** utilizes the data shown in FIG. **4**, and shows seven burst values, one burst value for each drop volume, plotted against the drop volume for a single liquid level (e.g. 3.01 mm in this case). In other words, FIG. **5** is generated by going to FIG. **4** and reading the burst value at 3.01 mm for each of the seven drop volumes. (In practice, this information would be extracted from the calibration files for the burst curves). In the preferred embodiment, the data in FIG. **5** are subjected to a curve fitting process, which in this case yields a straight line referred to as a calibration function **194**. Additionally, in the preferred embodiment two or more new calibration functions are generated in the same way that the calibration function **194** was generated, except that a new liquid level (and the corresponding new burst values) is used to generate each of the new calibration functions. For example, in FIG. **6** below, a total of thirty-six calibration functions were generated and subjected to a curve fitting process.

(29) The equation that results from the curve fitting process for the calibration function **194** is a linear equation that relates drop volume to burst value for a given liquid level. Therefore, the slope and intercept of the function **194** can be used to calculate the burst value needed to produce any selected drop volume at the specified liquid level. A least squares analysis of the data in FIG. **5**

yields a coefficient of determination ($R_{\text{sup.2}}$) of 0.9966, indicating a very good fit of the data to the straight line (calibration function **194**). In other embodiments, the calibration function may not be a straight line (i.e. the calibration function might be a curve), and a polynomial having a degree higher than one (e.g. 2-10) would be required for an acceptable the curve fit.

(30) FIG. **6** shows the slopes for thirty-six calibration functions plotted against liquid level. The thirty-six calibration functions were generated in the same way that the calibration function in FIG. **5** was generated. Specifically, the burst value for each of the seven drop volumes in FIG. **4**, is plotted against the drop volume for a single liquid level, calculating the slope and intercept of the resulting line, and then plotting the slope versus the liquid level to yield one of the data points in FIG. **6**. This is repeated thirty-five additional times to yield the results shown in FIG. **6**. A curve **200** may be drawn that connects all of the thirty-six data points in FIG. **6**. FIG. **7** shows the intercepts for the thirty-six calibration functions plotted against liquid level. A curve **210** may be drawn that connects all of the thirty-six data points in FIG. **7**.

(31) The data contained in FIGS. **6** and **7** can be used to calculate the burst value needed to produce any selected drop volume over the range of liquid levels. For example, Equation 1 can be used to calculate the required burst value:

Required burst=(slope)(desired drop vol.)+intercept (1) where the slope and intercept are obtained from FIGS. **6** and **7**, and the user measures the liquid level and chooses the desired drop volume.

(32) In a preferred embodiment, once a liquid level is measured and a desired drop volume has been selected, then the defined function for the point higher in the liquid level and the next point lower in the liquid level may be determined. The value for the actual point is determined by interpolation to the point measured and the proper burst value is acquired. For example, if the liquid level was measured at the point **202** on the curve **200**, then the slopes for points **204** and **206** in FIG. **6** would be determined, and the slope for point **202** would be determined by interpolation between these two burst values. Similarly, the intercept for the point **212** in FIG. **7** (at the measured liquid level) would be determined by interpolation between the points **214** and **216**. The interpolated values for the slope and intercept are then used in equation one to get the required burst for the new drop volume.

(33) FIGS. **6** and **7** show the results of plotting two coefficients, slope and intercept, for a linear calibration function, such as the calibration function **194** obtained in FIG. **5**. However, if the calibration function **194** was not a straight line, a higher degree polynomial would be required to fit the data to a curve. This higher degree polynomial would have additional coefficients that would be plotted in the same manner that the slope and intercept in FIGS. **6** and **7** were plotted.

(34) In a preferred embodiment, the present invention is implemented in software, so all of the data from the FIGS. **6** and **7** are stored in a lookup table in electronic memory. The lookup table lists liquid level, slope, intercept, higher degree coefficients (if any), and any other desired information (such as constants) in separate columns. Subsequently, an algorithm extracts the required data from the lookup table to yield the burst value needed to yield the specified drop volume at a given liquid level.

(35) A preferred embodiment of the method for using the present invention to dispense drops of source fluid **60** having any volume within a defined range is as follows: In a first step, a first burst curve data set **164** is created (e.g. using the apparatus **10**) that relates a range of liquid levels (FIG. **4**, x-axis) of a source liquid **40** to a range of burst values (FIG. **4**, y-axis) for dispensing one or more drops of the source liquid having a first drop volume (e.g. 2 nl), with the burst values being related to a plurality of acoustic waves **20**. In FIG. **4**, thirty-six reading at **36** liquid levels were used to create the burst curve **164**, so all of these data points are included in the term “first burst curve data set.” The phrase “burst values being related to a plurality of acoustic waves” means that burst is a series of acoustic waves followed by a period of rest.

(36) In a second step, a second burst curve data set **188** is also created that relates the range of

liquid levels to the range of burst values for dispensing one or more drops of the source liquid having a second drop volume (e.g., 10 nl), where the second drop volume is not equal to the first drop volume. More burst curve data sets (i.e., a plurality) could be created, such as the seven burst curves shown in FIG. 4, but two burst curve data sets are the minimum if the calibration function is going to be a straight line.

(37) In general, the defined range of drop volumes that can be dispensed using the present invention is approximately determined by the range of drop volumes used to create the burst curve data sets, which is 2 nl to 10 nl in this example. However, in other cases other ranges of drop volumes could be used. Frequently, the properties of the source solution being dispensed will influence the range of drop values selected. A preferred range of drop values is 1 nl to 25 nl. Additionally, in some cases, the defined range of drop volumes could be expanded outside of this range used to create the burst curve data sets, if the accuracy in the drop volumes produced outside of the range is acceptable.

(38) In a third step, a first calibration function data set **194** that relates the first drop volume (2 nl) to a first burst value measured at a first liquid level (3.01 mm in FIG. 5) in the first burst curve data set, and that relates the second drop volume (10 nl) to a second burst value measured at the first liquid level in the second burst curve data set.

(39) In a fourth step, a second calibration function data set is created that relates the first drop volume to a third burst value measured at a second liquid level in the first burst curve data set, and that relates the second drop volume to a fourth burst value measured at the second liquid level in the second burst curve data set. Here, the second liquid level is a liquid level not equal to the first liquid level. More calibration function data sets (i.e., a plurality) could be created, such as the thirty-six calibration function data sets used in FIG. 6, but two calibration function data sets are the minimum that would work in the present invention. The more calibration function data sets that are created, the better the usefulness of the invention.

(40) In a fifth step, a dispensing data set created from the first and second calibration function data sets is used to calculate a first new burst value required to dispense one or more drops of the source liquid having a first new drop volume, where the first new drop volume is different from both the first drop volume and the second drop volume. In a preferred embodiment, the first new burst value is calculated using the method described previously with respect to the points **202** and **212** in FIGS. 6 and 7, respectively. In the preferred embodiment, the first and second calibration function data sets are processed to yield two lines (i.e. two equations for lines), and the slopes and intercepts of these two lines are used in the dispensing data set. In other embodiments, the first and second calibration function data sets are processed to yield two polynomials having a degree higher than one, or some other type of equations that describe more complex curves, and the coefficients from these two polynomials, or from the other equations, are used in the dispensing data set.

(41) An important advantage of creating new volume calibrations on the fly is the ability to create a final dispense volume more efficiently (i.e. faster) by using the largest drop volume possible. In general, the most efficient method for achieving a final dispense volume (i.e. the total volume dispensed by a plurality of drops), is to use the largest drop volume that can be multiplied by an integer to yield the final dispense volume. In considering this issue, it should be recognized that minimum resolution for dispensing a drop is one burst. It is known that there are about 60 bursts per nanoliter of solution. This resolution corresponds to less than 2% of a one nanoliter dispense (i.e. 1/60 of a nanoliter is approximately 2%).

(42) To illustrate these advantages, FIG. 9 shows the drop volume required for a final dispense volume in the range of 0 to 100 nl, where the maximum drop size is 25 nl and the minimum drop size is 1 nl. The major point in FIG. 9 is that for any volume between 1 nl and 100 nl, a maximum of four drops is required to deliver the final dispense volume. For example, looking at 60 nl along the x-axis in FIG. 9, and reading up until the line **234** is intersected, shows that a drop volume of 20 nl is the required drop size ($3 \times 20 \text{ nl} = 60 \text{ nl}$).

(43) The information conveyed by FIG. 9 is important because it takes about 30 milliseconds to dispense a drop, so it would take about 2.94 seconds (98×30 msec) to dispense 99 nl of solution as 1 nl drops. In contrast, if 99 nl are dispensed using four drops of 24.75 nl, as can be done with the present invention, is only 0.09 seconds (3×30 msec). So, the method of the present invention improves (reduces) the time required to dispense a volume of solution, and also improves the resolution of the final dispense volume.

(44) FIG. 10 illustrates how a user of the acoustic dispensing apparatus 10 would use a preferred embodiment of the present invention to dispense a volume of source fluid. In step 244, the user specifies the maximum drop volume that can be used. In step 248, the user specifies the volume of source fluid that should be dispensed (in a single well). In step 252 the number of drops to be used to dispense the total volume from step 248 is calculated. In the preferred embodiment, Equation 2 is used for this, but other equations are acceptable. Additionally, the user can manually select and/or decide on the number of drops to be dispensed.

Total number of drops = integer(total volume / maximum drop volume) + 1 (2) In this Equation 2, the one extra drop is used to ensure that the maximum drop volume isn't exceeded.

(45) In step 256, the volume of the individual drops is automatically calculated, such as by dividing the volume to be dispensed by the number of drops. In step 260, the liquid level in the source well is automatically measured by the apparatus 10. In step 264, based on the liquid level from step 260, the coefficients for a calibration curve dependent on the liquid level are looked up, such as the slope and intercept coefficients from FIGS. 6 and 7. In step 268, the burst for the required drop volume is automatically calculated from the coefficients obtained in step 264, such as by using Equation 1: Required burst = (slope)(desired drop vol.) + intercept. Finally, in step 272 the required number of drops having the correct drop volume are dispensed by an acoustic dispensing apparatus, such as the apparatus 10.

(46) Although the present invention has been described in terms of the presently preferred embodiments, it is to be understood that such disclosure is not to be interpreted as limiting. Various alterations and modifications will no doubt become apparent to those skilled in the art after having read the above disclosure. Accordingly, it is intended that the appended claims be interpreted as covering all alterations and modifications as fall within the true spirit and scope of the invention.

Claims

1. A method for dispensing drops of a source liquid with a drop dispensing apparatus, the method comprising: specifying a maximum drop volume; specifying a total volume to be dispensed; calculating a number of drops to be dispensed; subsequent to specifying the maximum drop volume, calculating a particular drop volume required to dispense the total volume to reduce a dispense time required to dispense the total volume; measuring a liquid level of a solution to be dispensed; calculating a burst required to dispense the particular drop volume based at least in part on two or more liquid level dependent coefficients accessed from a dispensing data set; and using acoustic energy to dispense the number of drops required to yield the total volume to be dispensed.
2. The method of claim 1, wherein the two or more liquid level dependent coefficients include a slope value that gives a bursts per volume ratio used in calculating the particular drop volume.
3. The method of claim 1, wherein calculating the number of drops to be dispensed comprises: dividing the total volume to be dispensed by the maximum drop volume to produce a first result; and rounding the first result down to a nearest integer to produce a second result.
4. The method of claim 3, wherein calculating the number of drops to be dispensed further comprises adding one to the second result to produce the number of drops to be dispensed.
5. The method of claim 1, wherein calculating the particular drop volume comprises dividing the total volume to be dispensed by the number of drops to be dispensed.
6. The method of claim 1, wherein calculating the burst required to dispense the particular drop

volume comprises looking up the two or more liquid level dependent coefficients from a dispensing data set for the particular drop volume.

7. The method of claim 6, wherein the dispensing data set comprises a first coefficient of a first function corresponding to a first calibration function data set and a second coefficient of a second function corresponding to a second calibration function data set, wherein the first calibration function data set relates a first drop volume to a first burst value measured at a first liquid level and the second calibration function data set relates a second drop volume to a second burst value measured at a second liquid level.

8. The method of claim 6, wherein the dispensing data set comprises slopes and intercepts as a function of liquid level.

9. The method of claim 1, wherein the maximum drop volume is in a range of one nanoliter to twenty-five nanoliters.

10. A system comprising: an acoustic ejector configured to interface with a fluid reservoir and apply focused acoustic radiation thereto; and a controller comprising at least one processor and non-transitory memory containing instructions that, when executed by the processor, cause the controller to: receive a maximum drop volume; receive a total volume to be dispensed; determine a number of drops to be dispensed; subsequent to receive the maximum drop volume, determine a particular drop volume required to dispense the total volume to reduce a dispense time required to dispense the total volume; measure a liquid level of a solution to be dispensed; determine a burst of focused acoustic radiation from the acoustic ejector required to dispense the particular drop volume based at least in part on two or more liquid level dependent coefficients accessed from a dispensing data set; and cause the acoustic ejector to apply the burst of focused acoustic radiation to dispense the number of drops required to yield the total volume to be dispensed.

11. The system of claim 10, wherein the maximum drop volume is a range of drop volumes producible by the acoustic ejector and is in a range of one nanoliter to twenty-five nanoliters.

12. The system of claim 10, wherein the instructions to determine the particular drop volume comprises further instructions that, when executed by the processor, cause the controller to look up the two or more liquid level dependent coefficients from a dispensing data set for the particular drop volume.

13. The system of claim 12, wherein the dispensing data set comprises a first coefficient of a first function corresponding to a first calibration function data set and a second coefficient of a second function corresponding to a second calibration function data set, wherein the first calibration function data set relates a first drop volume to a first burst value measured at a first liquid level and the second calibration function data set relates a second drop volume to a second burst value measured at a second liquid level.

14. The system of claim 12, wherein the dispensing data set comprises slopes and intercepts as a function of liquid level.

15. The system of claim 10, wherein the two or more liquid level dependent coefficients include a slope value that gives a bursts per volume ratio used in calculating the particular drop volume.

16. The system of claim 10, wherein the instructions to determine the number of drops to be dispensed comprise further instructions that, when executed by the processor, cause the controller to: divide the total volume to be dispensed by the maximum drop volume to produce a first result; and round the first result down to a nearest integer to produce a second result.

17. The system of claim 16, wherein the instructions to determine the number of drops to be dispensed comprise further instructions that, when executed by the processor, cause the controller to add one to the second result to produce the number of drops to be dispensed.

18. The system of claim 10, wherein the instructions to determine the particular drop volume comprises further instructions that, when executed by the processor, cause the controller to further divide the total volume to be dispensed by the number of drops to be dispensed.
